

Modelling how children with Down syndrome learn language

Two studies: how vocabulary grows, and what reading-and-language teaching causes.



Study 01

Vocabulary growth

How children understand, say and sign words as they grow, across eleven international studies.



Study 02

Reading & language

A re-analysis of a randomised trial, separating what the teaching causes from what tracks ability.

One method, two studies

Both studies run on the same Bayesian foundation. What changes is the question, and how far the data can be pushed toward cause.

Study 01 · Vocabulary growth

Vocabulary growth

How do children build words?

Models the words children **understand, say and sign together**, pooled across eleven international studies. The comprehension–production gap falls straight out of the model as the production ratio **q**.

What’s distinctive

The understood × production-ratio decomposition, estimated jointly across eleven studies.

Study 02 · Reading & language

Reading & language

What does teaching cause?

Re-analyses a **randomised trial** through a locked causal DAG, so the teaching effect is identified by randomisation. Gradient-boosting discovery feeds the Bayesian estimates.

What’s distinctive

A causal DAG plus randomisation; a two-layer discovery-then-estimation method.

The shared foundation

Hierarchical Bayesian models

Beta–Binomial likelihood

Gaussian–process (HSGP) trends

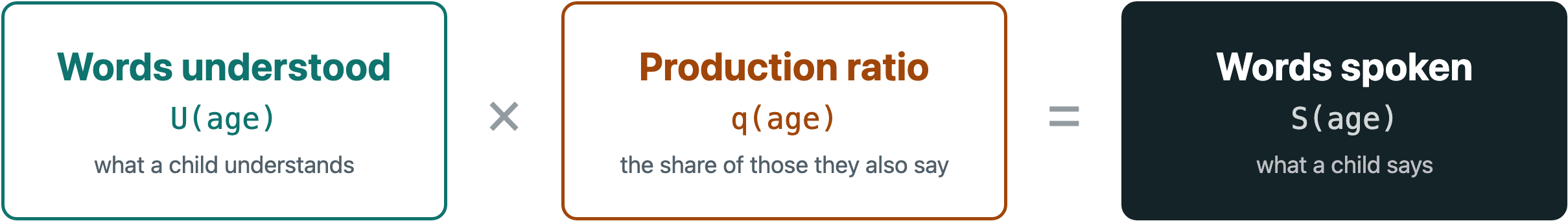
credible intervals throughout

reproducible: PyMC · NUTS · DuckDB · Quarto

The same engine underneath: **full-posterior inference**, checked with predictive checks and cross-validation, and reproducible from raw data to report.

How the vocabulary model works

The modelling idea at the centre of the study: spoken vocabulary is built from what a child understands and the share of it they say.



Modelling the two together means spoken vocabulary can **never exceed** what a child understands, and the comprehension-production gap falls out directly as the production ratio **q**.

The data

Parent-report vocabulary checklists, **pooled across 11 international studies** onto a common 810-word scale: **576 children**, ages 8 months to about 9½ years, roughly 1,000 observations.

The estimate



Each term is a **smooth curve over age** with full posterior uncertainty, so every result is a credible range, not a single number.

Hierarchical Beta-Binomial

Gaussian-process (HSGP) trends

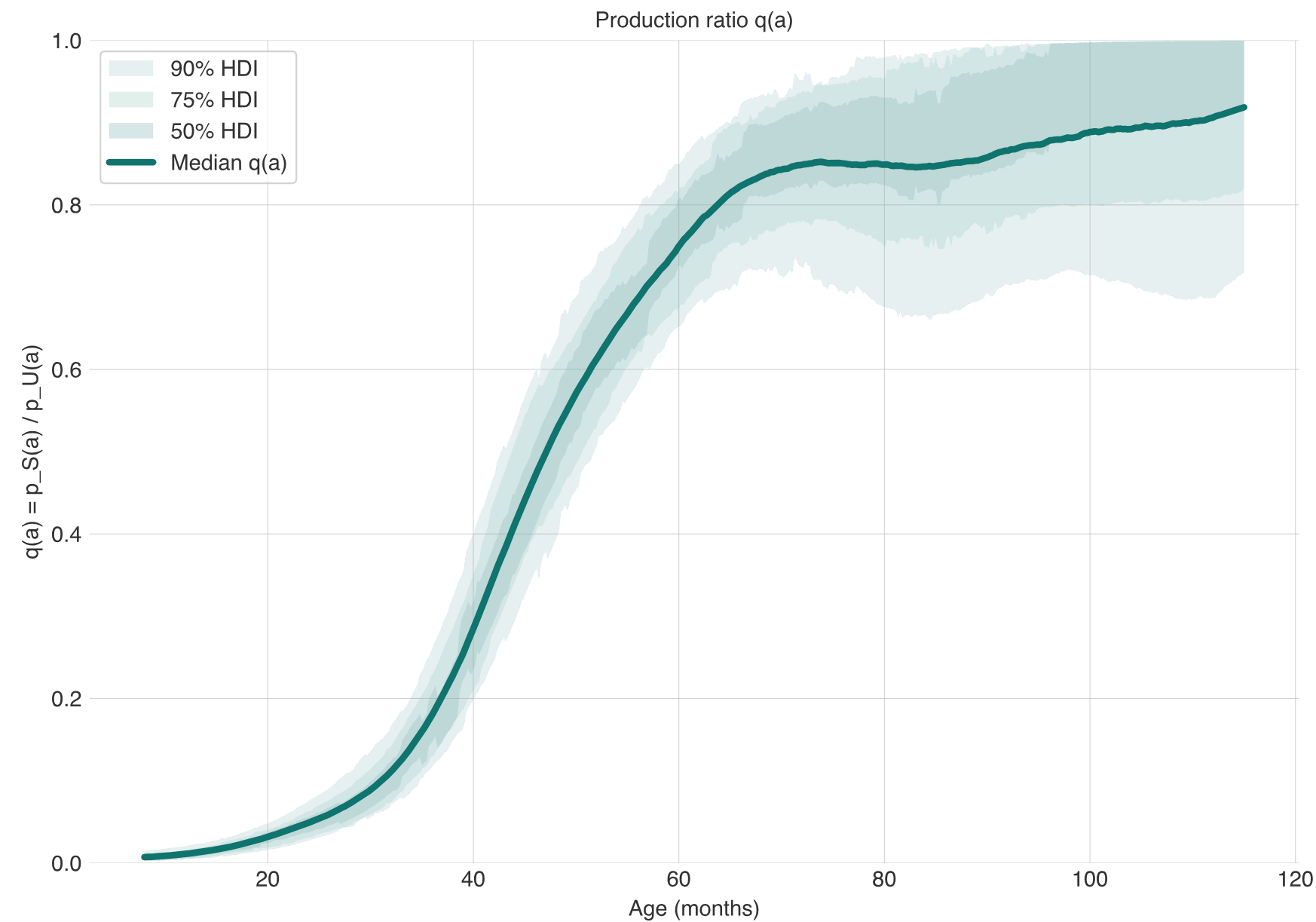
study & child random effects

PyMC / NUTS

PSIS-L00 comparison

Understanding runs years ahead of speaking

The share of understood words a child with Down syndrome also says, estimated across age from pooled international data.



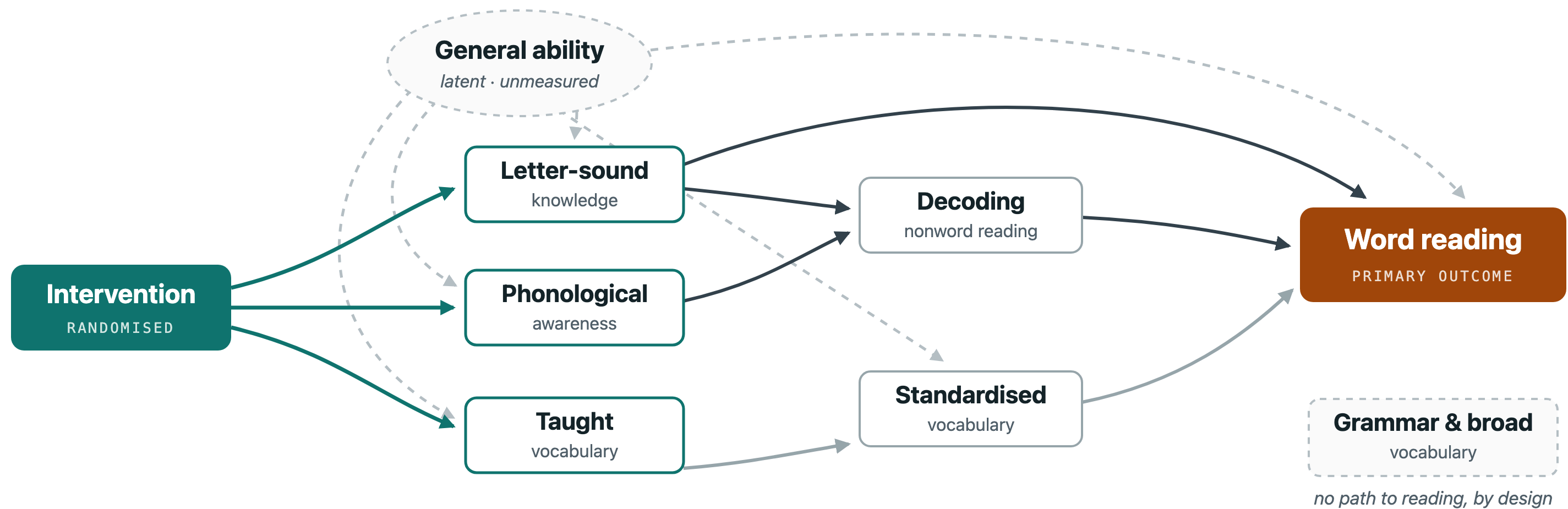
Each child's **production ratio** is the fraction of the words they understand that they also say. Comprehension leads production for years, so the ratio stays low at first, then climbs steeply.

~1% **~52%** **~86%**
at 1 year at 4 years by 7½ years

Wider than in typically-developing children at low vocabulary levels, narrowing as vocabulary grows.

A causal design for reading progress

Separating what the teaching *causes* from what merely correlates with a child’s general ability.



How to read it

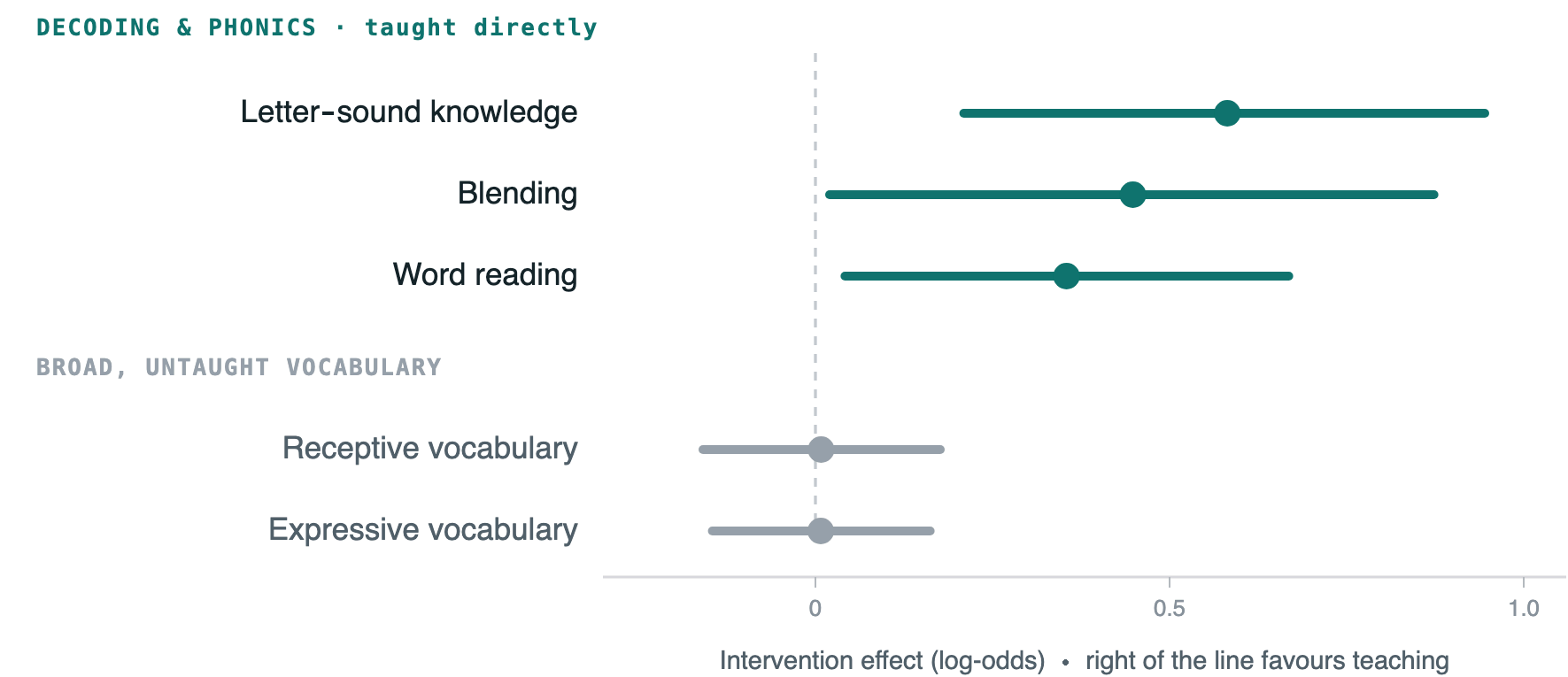
- Randomised effect, **causal**
- Hypothesised skill pathway, association
- General-ability confounding
- Association through standardised vocabulary
- Teal outline: a skill the teaching targets directly

The identification

Randomisation makes the intervention independent of general ability, so its effect on word reading is read as **genuine cause**. Every skill-to-skill link stays confounded by ability, so it is reported as **association, not cause**.

The intervention lifted reading, not broad vocabulary

The randomised effect of the teaching on each skill.



Each bar is the intervention’s effect on one skill, with its **95% credible interval**. Children were randomised, so these read as cause, not correlation.

The gains cluster on the skills taught most directly: letter-sounds, blending and word reading. Broad, untaught vocabulary barely moved.

Learning transfers near what’s taught, not far. The effect fades with distance from instruction. This is the transfer gradient.

● A bar clear of the dashed zero line is a credibly non-zero effect.